**MARK BARTON**

**C++ SOFTWARE ENGINEER & TECHNICAL LEADER · AUDIO SYSTEMS · DSP**

barkertrax@gmail.com · linkedin.com/in/mark-barton-b4282016

C++ software engineer and technical leader with 25+ years' experience developing real-time audio software, DSP and cross-platform systems. Combines hands-on implementation and debugging with architectural direction and engineering leadership, spanning consumer electronics, gaming and PC audio.

## TECHNICAL SKILLS

- **Languages:** C++, C, C#, Python
- **Audio:** DSP algorithms, cross-platform audio SDKs
- **Audio platforms:** Windows Audio Processing Objects, Sound Open Firmware, SoC DSP integration, VST3
- **Platforms:** Windows, macOS, iOS, Android, ChromeOS, Linux

## PROFESSIONAL EXPERIENCE

### DTS / Xperi | Senior Manager, PC Audio

*2023 to Present*

- Own the implementation and evolution of a scalable PC audio architecture spanning SoC-based DSP processing, Windows Audio Processing Objects and end-user applications.
- Continue hands-on development in the production C++ codebase, implementing and debugging audio software as part of ongoing architectural development.
- Lead two UK-based teams, with responsibility for the development and performance management of four direct reports and leadership of three indirect reports.
- Encourage practical adoption of AI-assisted development and a team-specific AI skills framework aligned with the codebase, engineering workflows and CI.

### DTS / Xperi | Staff Software Engineer, PC Audio

*2019 to 2023*

- Provided technical leadership for audio post-processing technologies deployed on consumer PC platforms.
- Developed an XML-driven system for defining audio processing modules and automatically generating C++ implementations, tests and VST3 components, improving consistency and reducing repetitive development work.
- Consolidated multiple continuous integration systems into a unified CI system supporting the end-to-end PC software ecosystem.

- Implemented DSP algorithms and audio processing SDKs across PC, macOS, iOS and Android, including integration with DSP platforms from Intel, AMD and NVIDIA.
- Primary developer of a ChromeOS post-processing solution integrating via Sound Open Firmware (SOF), working across kernel-facing and DSP-facing interfaces on Intel, AMD and MediaTek SoCs. The solution shipped via ChromeOS updates using Google's standard release mechanisms.

**Electric Square | Audio Programmer**
*2017 to 2019*

- Primary Audio Programmer on *Forza Street* and *Hot Wheels ID*.

**Mytrax | Audio Programmer**
*2014 to 2016*

- Worked on IntraWave, using high-frequency audio signals for indoor positioning.
- Developed and shipped an iOS and Android application that applied the audio characteristics of Japanese live venues to a music library in real time.

**SELECTED EARLIER ENGINEERING EXPERIENCE**

- **Sony Computer Entertainment**: Implemented a real-time procedural audio synthesis system for PlayStation 3, targeting both the PPU and SPU.
- **Pivotal Games**: Designed a dynamic audio environment system for games and developed a cross-platform C++ Voice over IP library for online multiplayer. Worked on multiple commercially successful, multi-platform game releases.
- **Tsugi Studio**: Developed a C++/CLI library for hosting VST-compatible audio plugins, audio processing tools and a .NET library supporting multiple audio formats.

**EDUCATION**

**BEng Electronics, Music and Media Technology, First Class Honours**
Leeds Beckett University

**Certificate of Higher Education, Music Technology, Distinction**
London Metropolitan University

**PERSONAL PROJECTS**

- fromewardbound.co.uk: Documentation of a 25,000 km overland journey from Japan to the UK in a 1989 Toyota Land Cruiser.
- barkertrax.com: Nascent website documenting ongoing personal projects.
- Home renovation: Major renovation of a 1970s bungalow, including substantial demolition and the construction of a modernist home around the retained structure.